

RESEARCH ARTICLE

Thermodynamic equilibrium analysis of steam methane reforming based on a conjugate solution of material balance and law action mass equations with the detailed energy balance

Dmitry Pashchenko 

Samara State Technical University,
Molodogvardeiskaya str. 244, Samara
443100, Russia

Correspondence

Dmitry Pashchenko, Samara State
Technical University, Molodogvardeiskaya
str. 244, Samara 443100, Russia
Email: pashchenkodmitry@mail.ru

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Summary

Thermodynamic equilibrium analysis of the steam methane reforming (SMR) process to synthesis gas was studied. For this purpose, the system equations of the material balance and the equations of law mass action were solved by dichotomy method. The investigation was performed for a wide range of operational conditions such as a temperature, pressure, and inlet steam-to-methane ratio. The results obtained, with the help of developed algorithms, were compared with the results obtained via different commercial and open-source programs. All results are in excellent agreement. The operational conditions for the probable formation of carbon were determined. It was established that for the temperature range above 1100 K, the probability of carbon formation is absent for steam-to-methane ratio above units. In order to determine the amount of heat supplied per 1 mol to the reformer, the heat balance equation was obtained to achieve a targeted degree of methane conversion. With the help of the heat balance equation, it was established that the resulting transformation of substances in the steam methane reformer can be presented as a sequential heating of feed streams of methane and steam from the inlet temperature T_1 to the outlet temperature T_2 , heat for SMR reaction at the temperature T_2 , and heat for transformation of the part of the produced carbon monoxide (CO) via the water-gas shift (WGS) reaction at the temperature T_2 . The presented algorithm of thermodynamic analysis gives an appearance of the dependence of the product composition and the amount of required heat from operating conditions such as the temperature, pressure, and steam-to-methane ratio.

KEYWORDS

heat balance, methane, steam methane reforming, thermodynamic analysis

1 | INTRODUCTION

Steam methane reforming (SMR) is one of the vital methods for production of hydrogen, carbon monoxide, or synthesis gas from natural gas that mainly consists of

methane.^{1,2} This is achieved in the processing setups that reacts steam at high temperature with methane. This setup is known as the steam methane reformer. The steam methane reformer is also widely used in fuel cells for hydrogen production.^{3,4} There is also interest in using the

SMR reaction for thermochemical recuperation of waste flue gases heat.^{5,6}

It is generally known that the efficiency of SMR process significantly depends on operational parameters such as temperature, composition of the inlet reaction mixture, and pressure.⁷ In addition, the composition of reformed products (reformat) in real reformers also depends on the type of catalyst, the mass of catalyst, and the form and method of filling the reaction space.^{8,9} However, it is known that modern high-performance catalysts aid in retrieving the composition of reformed products at the outlet, which is close to the equilibrium.^{10,11} Therefore, researchers and engineers pay significant attention to the determination of equilibrium compositions (maximally achievable concentrations) of the reformed products against of technological parameters: pressure, inlet steam-to-methane ratio, and temperature.

The thermodynamic analysis allows to calculate the effects of temperature, pressure, and inlet gas composition on equilibrium reformat compositions. In recent times, the results of thermodynamic equilibrium analysis by using the Gibbs free energy minimization (GFEM) method for SMR have been published by different specialists.^{7,12,13} The wide development of specialized software significantly simplified the implementation of thermodynamic analysis of SMR. The researcher uses quite literally just a few mouse clicks away in setting operational parameters (temperature, pressure, and feed gas composition) and after a few seconds is getting the result. The different commercial and open-source software is used for the prediction and modeling of SMR process. For example, Aspen-HYSYS,¹⁴ IVTANTHERMO,¹⁵ CHEMCAD,¹⁶ Chemkin, and ProVision are all programs that are using different databases and algorithms to calculate the equilibrium gas composition.

Özkara-Aydınoğlu presented the results of thermodynamic equilibrium analysis of the methane reforming process with carbon dioxide (CO₂) and steam (H₂O) to produce the mixture of hydrogen and carbon monoxide.⁷ The effect of pressure, various inlet gas compositions, and the temperatures on the reformat composition was performed with the help of GFEM technique by using Aspen-HYSYS software. This software is capable of modeling a single phase or multiphase of multicomponents in equilibrium conditions. The Gibbs reactor in Aspen-HYSYS is used to determine reformat compositions in the steam methane reformer under the conditions of GFEM. With Aspen-HYSYS, the composition of the reforming process is calculated by GFEM of each elements in the reaction system according to the reaction networks.

In the author's previous works for the thermodynamic analysis of propane and methane reforming, the program IVTANTHERMO was used.^{14,17} The main difference of

this software is the use of different databases for the process modeling. At the same time, the results obtained with Aspen and IVTANTHERMO for propane reforming are almost identical.

Sun et al¹⁸ presented thermodynamic study of SMR process via FactSage software. The equilibrium compositions are determined by minimizing the Gibbs free energy for a given set of species without any specification of the possible reactions, which might take place in the steam methane reformer. The GFEM method was realized via special software FactSage.

In addition, Tabrizi et al¹⁹ presented the results of thermodynamic analysis of the SMR process with statistical approaches. The authors investigated the SMR process via combination of thermodynamic analysis by GFEM method and statistical modeling with the full quadratic polynomial equations. In that work, the authors obtained the mathematical relationships between pressure, temperature, and steam-to-methane ratio were approximately determined by the second-order polynomial. Moreover, there are the articles where the SMR process was investigated via combination of thermodynamic analysis and regression analysis.²⁰ Fuqiang et al²⁰ investigated the effect of all independent variables on the equilibrium outlet gas composition when regression analysis is adopted for examining the quantitative relationship between response observed and conditions.

The works mentioned above consider the algorithm for calculating the equilibrium composition as a "black box" in which it is difficult to establish the effect of operational parameters on the result of the calculation. In addition, it is difficult for the starting investigators to see the relationship between the calculation results and the initial data. One of the simplest and obvious ways to perform a thermodynamic analysis of methane steam reforming is to conjugate solve the integral material balance equations and the mass action law equations for the SMR process. The use of this method makes it possible to obtain a visual relationship between the initial data and the calculation results. Moreover, the conjugate solution of the integral material balance equations and the mass action law equations allows to calculate the boundary conditions for the carbon formation in the steam reformer and to calculate heat balance for the SMR process.

The main goals of the present work are

- to develop and describe the algorithm of thermodynamic analysis based on the conjugate solution of the integral material balance equations and the mass action law equations;
- to analyze the chemical reactions in the SMR system; and

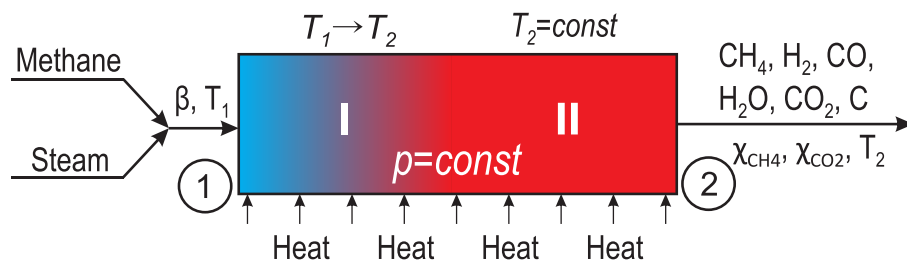


FIGURE 1 The flow diagram for substance flows in the steam methane reformer

TABLE 1 Possible chemical reactions in the reformer⁸

No.	Reaction	ΔH_{298} , kJ/mol
Methane oxidation		
1	$\text{CH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CO} + 3\text{H}_2$	206.4
2	$\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$	-41.1
3	$\text{CH}_4 + \text{CO}_2 \rightleftharpoons 2\text{H}_2 + 2\text{CO}$	247.3
Carbon formation		
4	$\text{CH}_4 \rightleftharpoons \text{C} + 2\text{H}_2$	74.8
5	$2\text{CO} \rightleftharpoons \text{C} + \text{CO}_2$	-173.3
6	$\text{CO} + \text{H}_2 \rightleftharpoons \text{C} + \text{H}_2\text{O}$	-90.13
7	$\text{CO}_2 + 2\text{H}_2 \rightleftharpoons \text{C} + 2\text{H}_2\text{O}$	-131.3

- to calculate and verify the results of thermodynamic analysis.

2 | CHEMICAL MECHANISM

The flow diagram for substance flows in the steam methane reformer is shown in Figure 1. For analyzing the process, it is further assumed that a mixture of pure methane and pure steam is supplied to the reformer. For more accurate calculations, it is necessary to take into account the presence of impurities of hydrocarbons, nitrogen, and other components in the gas flow.

There are a lot of reactions in the steam methane reformer to produce hydrogen (H_2), carbon (C), carbon monoxide (CO), and carbon dioxide (CO_2), and other species may include the reaction presented in Table 1.⁸ Moreover, other chemical reactions can take place in the steam methane reformer. These reactions are mainly a combination of the equations listed in Table 1. To perform a thermodynamic analysis of SMR, it is necessary and sufficient to analyze only the elementary chemical reactions listed in Table 1.

Based on GFEM method, from the several possible chemical reactions that occur in the reformer the most probable reactions are reactions with a minimum change in Gibbs free energy. There are many articles that consider the algorithm for the determination of Gibbs free energy.^{7,17}

For a system at chemical equilibrium, the Gibbs free energy for each reaction can be determined as follows:

$$\Delta G^\circ = -RT \ln K_{eq(i)}, \quad (1)$$

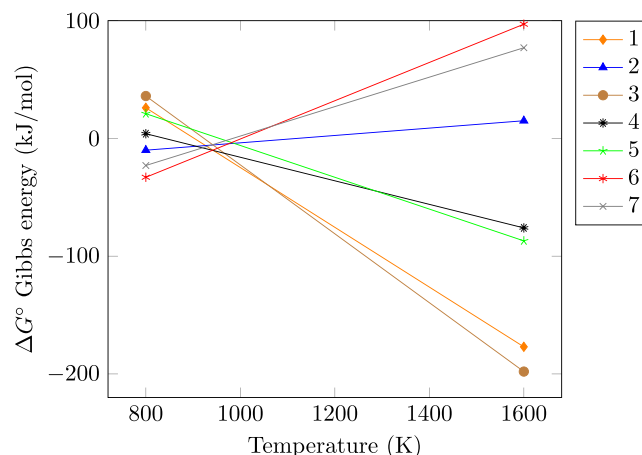


FIGURE 2 The Gibbs free energy as a function of temperature for reactions in Table 1

where R is gas constant, T is temperature (for analyzed scheme, T is equal to T_2 according to Figure 1), and $K_{eq(i)}$ is equilibrium constant of i -reaction.

Figure 2 shows the dependence of the Gibbs free energy versus temperature for reactions presented in Table 1.

Figure 2 shows that change in Gibbs free energy of chemical reactions 1 and 3 in Table 1 is decreasing notably when temperature in the reformer increases. When temperature in the reformer is over 900 K, reactions 1 and 3 in Table 1 are most probable from a thermodynamic point of view than reactions 4 and 5. For the temperature range over 1000 K, the equilibrium conditions for reactions 6 and 7 is moved to the side of carbon monoxide and hydrogen production while carbon is consumed.

3 | METHODOLOGY

The material balance equation for the chemical-technological system (Figure 1) for the steady-state condition can be transcribed as

$$\sum_{j=1}^l \dot{n}_{i,j} = 0, \quad (2)$$

where $\dot{n}_{i,j}$ is the molar flow rate of i -element through j inlet (or outlet) (mol/s), l is the total number of inlets and out-

lets, the flows to the reformer is considered as positive, and the flows from the reformer are negative.

For further analysis at the reformer inlet $j = 1$, at the reformer outlet $j = 2$. Then the equation of material balance for carbon takes the form

$$\dot{n}_{CH_4,1} - \dot{n}_{CH_4,2} - \dot{n}_{CO_2} - \dot{n}_{CO_2,2} = 0. \quad (3)$$

The equation of material balance for oxygen:

$$\dot{n}_{H_2O,1} - \dot{n}_{H_2O,2} - \dot{n}_{CO_2} - 2\dot{n}_{CO_2,2} = 0 \quad (4)$$

The equation of material balance for hydrogen:

$$4\dot{n}_{CH_4,1} + 2\dot{n}_{H_2O,1} - 4\dot{n}_{CH_4,2} - 2\dot{n}_{H_2,2} - 2\dot{n}_{H_2O,2} = 0. \quad (5)$$

The material balance equations (3-5) allow to minimize the number of parameters describing the resulting change in the composition of the reaction mixture at the reformer outlet. This significantly facilitates the subsequent analysis of the functioning of the system.

The inlet gas mixture composition can be unambiguously determined as

$$\beta = \frac{\dot{n}_{CH_4,1}}{\dot{n}_{H_2O,1}}. \quad (6)$$

The reformat composition can be determined by two parameters:

$$X_{CH_4} = \frac{\dot{n}_{CH_4,1} - \dot{n}_{CH_4,2}}{\dot{n}_{CH_4,1}}; X_{CO_2} = \frac{\dot{n}_{CO_2,2}}{\dot{n}_{CH_4,1}}. \quad (7)$$

The first parameter is the methane conversion degree. The second is the degree of methane conversion to carbon dioxide.

Therefore, the composition of the outlet products can be expressed using parameters β , X_{CH_4} , and X_{CO_2} .

Methane:

$$\dot{n}_{CH_4,2} = \dot{n}_{CH_4,1}(1 - X_{CH_4}); \quad (8)$$

Carbon monoxide:

$$\dot{n}_{CO_2} = \dot{n}_{CH_4,1}(X_{CH_4} - X_{CO_2}); \quad (9)$$

Steam:

$$\dot{n}_{H_2O,2} = \dot{n}_{CH_4,1}(\beta - X_{CH_4} - X_{CO_2}); \quad (10)$$

Hydrogen:

$$\dot{n}_{H_2,2} = \dot{n}_{CH_4,1}(3X_{CH_4} + X_{CO_2}). \quad (11)$$

For these equations, $X_{CO_2} < X_{CH_4} < 1$ and $\beta > (X_{CO_2} + X_{CH_4})$.

Assuming that at sufficiently high temperatures maintained during SMR, the reagents exhibit the properties of ideal gases (in the sense that the chemical potentials of the

components are determined by their partial pressures), the equilibrium constant for water-gas shift (WGS) reaction can be written as

$$K_{eq,wgs} = \frac{P_{H_2} \cdot P_{CO_2}}{P_{CO} \cdot P_{H_2O}}. \quad (12)$$

The equilibrium constant (Equation 12) can be expressed by molar fraction of each species:

$$K_{eq,wgs} = \frac{x_{H_2} \cdot x_{CO_2}}{x_{CO} \cdot x_{H_2O}}, \quad (13)$$

where $x_i = \frac{\dot{n}_{i,2}}{\sum \dot{n}_{i,2}}$. Moreover, here, $\sum \dot{n}_{i,2} = \dot{n}_{CH_4,1}(1 + 2X_{CH_4} + \beta)$.

Therefore, from Equation (13), on the basis of expressions (9) to (11), the following equation can be obtained:

$$\begin{aligned} &\dot{n}_{CH_4,1}^2 (3X_{CH_4} + X_{CO_2}) \cdot X_{CH_4} \\ &- K_{eq,wgs} \cdot \dot{n}_{CH_4,1}^2 (X_{CH_4} - X_{CO_2}) (\beta - X_{CH_4} - X_{CO_2}) = 0. \end{aligned} \quad (14)$$

This equation can be represented as a quadratic equation:

$$a \cdot X_{CO_2}^2 + b \cdot X_{CO_2} + c = 0, \quad (15)$$

where $a = 1 - K_{eq,wgs}$, $b = 3X_{CH_4} + K_{eq,wgs} \cdot \beta$, $c = -K_{eq,wgs} \cdot X_{CH_4} \cdot (\beta - X_{CH_4})$.

In Equation (15), $b > 0$ and $c < 0$ because $\beta > X_{CH_4}$. Equation (15) has two roots:

$$X_{CO_2} = \frac{-b - \sqrt{b^2 - 4ac}}{2a}; \quad X_{CO_2} = \frac{-b + \sqrt{b^2 - 4ac}}{2a}. \quad (16)$$

If $K_{eq,wgs} > 1$ (at the temperature range below 1090K), then $a < 0$, and therefore, $\sqrt{b^2 - 4ac} > b$ and both roots of Equation (15) are positive. However the root with “-” before $\sqrt{b^2 - 4ac}$ does not satisfy condition $X_{CO_2} < X_{CH_4}$, due to the fact that

$$\frac{-b - \sqrt{b^2 - 4ac}}{2a} > \frac{b}{|a|} > \frac{K_{eq,wgs} \cdot \beta}{K_{eq,wgs} - 1} > \beta > X_{CH_4}. \quad (17)$$

At high temperatures, ($T > 1100$ K) $K_{eq,wgs} < 1$; therefore, $a > 0$, and the root with “-” before $\sqrt{b^2 - 4ac}$ also does not satisfy, since in this case it is less than 0. Hence, it follows that there exists a unique solution of Equation (15) that has a physical meaning:

$$X_{CO_2} = \frac{-b + \sqrt{b^2 - 4ac}}{2a}. \quad (18)$$

Equation (15) allows us to exclude X_{CO_2} as the independent parameter of the system, associating it with the parameters β , X_{CH_4} , and T_2 (by $K_{eq,wgs}$).

According to Figure 1, a constant temperature T_2 is maintained at a sufficiently large area near the reformer outlet. It should be expected that in the reacting mixture, an equilibrium composition of all the elements will be taking place.

In order to determine the degree of methane conversion, it is necessary to solve, with respect to X_{CH_4} , an equation describing the equilibrium with respect to the first reaction (according to Table 1) (equilibrium in the WGS reaction is already taken into account in expression 15):

$$K_{eq,smr} = \frac{P_{H_2}^3 \cdot P_{CO}}{P_{CH_4} \cdot P_{H_2O}} = \frac{x_{H_2}^3 \cdot x_{CO}}{x_{CH_4} \cdot x_{H_2O}} \cdot P^2, \quad (19)$$

where P is the total pressure in the reformer.

Equation (19) implicitly determines the required dependence X_{CH_4} versus T_2 . The solution of this equation with respect to X_{CH_4} in view of its nonlinearity and the dependence of the right-hand side on T_2 can be carried out by the different numerical methods. For this thermodynamic analysis, the dichotomy method was used to determine the degree of methane conversion.²¹ According to the definition, the dichotomy method is a root-finding method that applies to any continuous functions for which one knows two values with opposite signs. The method consists of repeatedly bisecting the interval defined by these

values and then selecting the subinterval in which the function changes sign and therefore must contain a root.²²

Figure 3 shows the equilibrium changes of the degree of methane conversion as a function of temperature when the different inlet gas compositions are introduced into the SMR reaction system at the various pressure. Steam additions have a noteworthy effect on the degree of methane conversion. The steam addition to the inlet reaction mixture is increasing the methane conversion notably, especially in the temperature range before 1200 K. This facts are correlated with the works of other authors.⁷ Özkara-Aydınoglu confirmed that H_2O addition in the inlet reaction mixture is increasing the methane conversion. Above 1200 K, the degree of methane conversion is obtained about maximum value (about 100%). This fact also has a worthy correlation with the articles of other authors.¹⁸

The trends in Figure 3 indicates that a decrease in pressure increases the methane conversion. This is a consequence of the Le-Chatelier principle. This fact can be explained by Equation (19) where right side of the equation is concluding the total pressure in the reformer.

To verify this developed algorithm, the results were compared with the results that were obtained via the commercial and open-source softwares: Aspen-HYSYS and IVTANTHERMO (Table 2). All results are in excellent

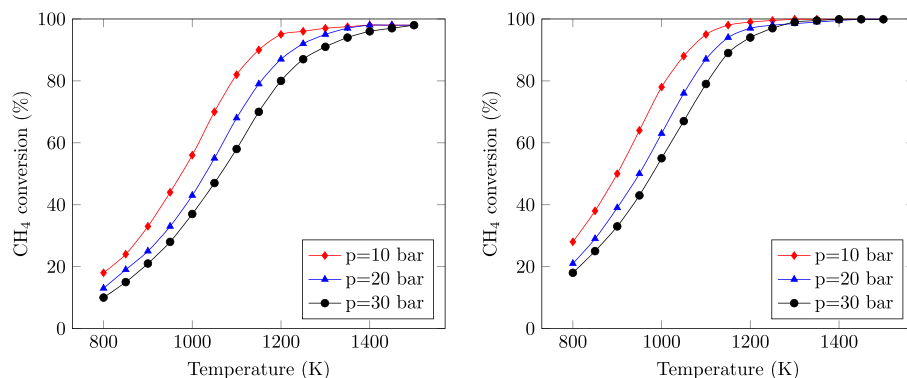


FIGURE 3 Effect of temperature on the equilibrium degree of methane conversion for the various pressure: left $\beta = 2$; right $\beta = 4$

Temperature, K	Pressure, bar	Source	Gas Composition, %				
			H ₂	CO	H ₂ O	CO ₂	CH ₄
1000	1	IVTAN	63.01	14.32	16.32	5.23	1.12
		ASPEN	63.40	14.38	16.12	5.08	1.02
		Algorithm	63.42	14.43	16.11	5.12	0.92
1000	5	IVTAN	60.66	14.82	17.32	5.55	1.65
		ASPEN	61.60	14.98	17.92	4.08	1.42
		Algorithm	61.44	14.93	17.97	4.18	1.48
1100	1	IVTAN	63.88	16.08	16.02	3.93	0.09
		ASPEN	63.65	16.14	16.15	3.98	0.08
		Algorithm	63.76	16.11	16.11	3.94	0.08
1100	5	IVTAN	61.28	15.02	17.87	4.24	1.59
		ASPEN	61.57	14.99	17.88	4.11	1.45
		Algorithm	61.45	14.93	17.96	4.18	1.48

TABLE 2 The comparison of the reformate composition, % (mol/mol)

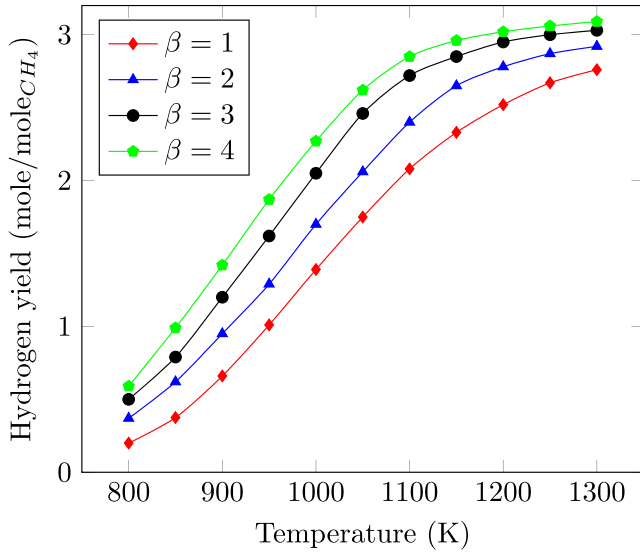


FIGURE 4 Hydrogen yield for the various β ratio at $p = 10$ bar

agreement with the simulated data obtained via these special programs.

With the help of the material balance equations, it is possible to determine absolute volume of hydrogen at the reformer outlet. Figure 4 shows that the number of moles of hydrogen increases when temperature and β ratio increasing. It was established that for the high temperature, more than 3 mol of hydrogen can be produced for each mole of methane, while the equilibrium composition for SMR reaction (reaction 1 according to Table 1 gives only 3 mol). Other hydrogen is produced by WGS reaction. That is why the WGS reaction must be taken into account for the thermodynamic analysis of the SMR process.

Figure 5 displays equilibrium composition of gas mixture at the reformer outlet against the temperature in the steam methane reformer for the steam-to-methane ratio $\beta = 2$ and pressure $p = 10$ bar. In this figure, CH_4 conversion and the hydrogen-to-carbon monoxide outlet ratio (H_2/CO) is increasing when the reformer temperature increases. As detailed above, a simple and intuitive algorithm of thermodynamic analysis makes it possible to determine the equilibrium gas composition at the outlet of the steam methane reformer without an expensive commercial software. In addition, the developed algorithm shows the apparent dependence of operational parameters, such as temperature, composition of the initial mixture, and pressure on the equilibrium composition.

There are some hints that significantly simplify thermodynamic analysis in the temperature range above 1200 K (usually steam methane conversion is carried out at temperatures above 1100–1200 K) and in the temperature range below 700 K.

For the temperature range below 700 K, $K_{eq,smr} \ll 1$ and $K_{eq,wgs} > 1$, which suggests $X_{\text{CH}_4} \ll 1$ and also $X_{\text{CH}_4} \sim$

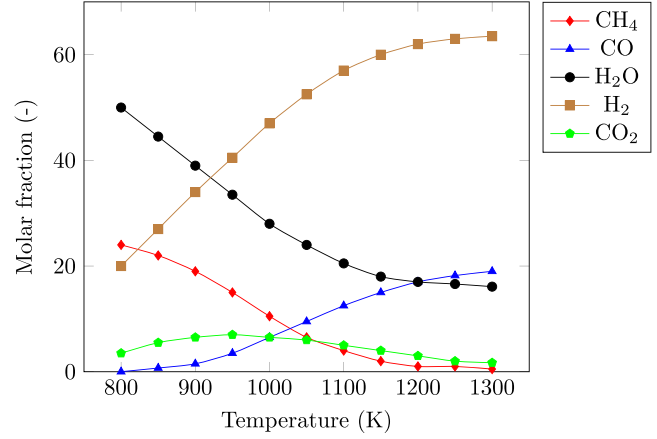


FIGURE 5 The equilibrium gas composition for $\beta = 2$ (solid line) and $\beta = 4$ (dashed line) at $p = 10$ bar

X_{CO_2} . Therefore, Equations (13) and (19) can be simplified as

$$K_{eq,smr} = \frac{(4X_{\text{CH}_4})^3}{(1 - X_{\text{CH}_4})(\beta - 2X_{\text{CH}_4})(1 + 2X_{\text{CH}_4} + \beta)^2}; \quad (20)$$

$$K_{eq,wgs} = \frac{(4X_{\text{CH}_4})(X_{\text{CH}_4})}{(\beta - 2X_{\text{CH}_4})}. \quad (21)$$

If we multiply Equation (20) and Equation (21), let $X_{\text{CH}_4} \ll 1$ and $\beta = 1$, then we obtain the simple equation:

$$K_{eq,smr} \cdot K_{eq,wgs} = 64 \cdot X_{\text{CH}_4}^5. \quad (22)$$

For the temperature range above 1200 K, $K_{eq,smr} \gg 1$ and $K_{eq,wgs} \ll 1$, which suggests $X_{\text{CH}_4} \sim 1$ and also $X_{\text{CO}_2} \sim 0$. Therefore, Equations (13) and (19) can be simplified as

$$K_{eq,smr} = \frac{(3X_{\text{CH}_4}^3)(X_{\text{CH}_4})}{(1 - X_{\text{CH}_4})(\beta - X_{\text{CH}_4})(1 + 2X_{\text{CH}_4} + \beta)^2}; \quad (23)$$

$$K_{eq,wgs} = \frac{(3X_{\text{CH}_4})}{(X_{\text{CH}_4})(\beta - X_{\text{CH}_4})}. \quad (24)$$

If we multiply Equation (23) and Equation (24), let $X_{\text{CH}_4} \sim 1$ and $\beta = 1$, then we obtain the simple equation:

$$K_{eq,smr} \cdot K_{eq,wgs} = 4.5 \frac{X_{\text{CH}_4}^4}{(1 - X_{\text{CH}_4})(1 - X_{\text{CH}_4}^2)}. \quad (25)$$

To verify the results obtained via the presented algorithm, error analysis was performed. The methane conversion obtained using the presented algorithm was compared with the methane conversion obtained using Aspen-HYSYS software.^{15,23} An error distribution histogram for the methane conversion is presented in Figure 6. Figure 6 depicts that the best possible fit between results was obtained for the steam-to-methane

ratio above stoichiometric ($\beta > 1$). This fact can be explained as follows with a stoichiometric steam-to-methane ratio at the reformer inlet, side reactions from Table 1 occur. Moreover, with $\beta = 2$ and $\beta = 3$, the SMR and WGS reactions mainly occur in the reformer.

The error analysis with the help of the Pearson criterion, accomplished on the 9-bit histogram basis in Figure 6, presented that the error distributions for the methane conversion at pressure of 1 bar and temperature of 900 K within the steam-to-methane ratio of 2 are well correlated with the law of the normal Gaussian distribution ($\chi_{func} = 3.77 < \chi_{aspen} = 6.18$, with a confidence level of $P = 92\%$ and the number of degrees of freedom $m = 6$). Therefore, for the engineering calculations with high accuracy, it is possible to use the shown algorithm to determine the equilibrium gas composition at the reformer outlet.

4 | CARBON FORMATION

Carbon formation is another aspect that is important to consider by thermodynamic analysis.^{24,25} For many publications devoted to thermodynamic analysis of SMR, the carbon formation was omitted. We can predict favorable conditions for carbon formation more accurately by utilizing the developed algorithm. From the analysis of the chemical equations shown in Table 1, it is seen that in order to prevent the carbon formation, the following conditions must be fulfilled:

for temperature above approximately 950 K,

$$\frac{P_{H_2}^2}{P_{CH_4}} > K_{eq,4}; \quad \frac{P_{CO_2}}{P_{CO}^2} > K_{eq,5}; \quad (26)$$

for temperature below approximately 950 K,

$$\frac{P_{CO} \cdot P_{H_2}}{P_{H_2O}} < K_{eq,6}; \quad \frac{P_{CO_2} \cdot P_{H_2}^2}{P_{H_2O}^2} < K_{eq,7}; \quad (27)$$

where i -index for the equilibrium constants $K_{eq,i}$ is number of reaction in Table 1.

To determine the equilibrium composition, we must write the material balance equation and the law mass equation for the temperature range above approximately 950 K and below approximately 950 K as it was shown above. For equation system with eight equations, we will have six variables: $\dot{n}_{CH_4,2}$, $\dot{n}_{CO_2,2}$, $\dot{n}_{H_2,2}$, $\dot{n}_{H_2O,2}$, $\dot{n}_{CO,2}$, and β .

$$\begin{cases} \dot{n}_{CH_4,2} = \dot{n}_{CH_4,1}(1 - X_{CH_4}) \\ \dot{n}_{CO,2} = \dot{n}_{CH_4,1}(X_{CH_4} - X_{CO_2}) \\ \dot{n}_{H_2O,2} = \dot{n}_{CH_4,1}(\beta - X_{CH_4} - X_{CO_2}) \\ \dot{n}_{H_2,2} = \dot{n}_{CH_4,1}(3X_{CH_4} + X_{CO_2}) \\ K_{eq,4} > \frac{x_{H_2}^2}{x_{CH_4}} \cdot P; K_{eq,6} > \frac{x_{CO_2}}{x_{CO}^2} \cdot \frac{1}{P} \\ K_{eq,6} > \frac{x_{CO} \cdot x_{H_2}}{x_{H_2O}} \cdot P; K_{eq,7} > \frac{x_{CO_2} \cdot x_{H_2}^2}{x_{H_2O}^2} \cdot P \end{cases}$$

A system of equations consisting of the equations of material balance and the law mass equation can be solved by any numerical method. An iterative method of successive approximations was used for this equation system.

Figure 7 depicts the minimum temperature for the various β ratio at which the steam methane reformer can operate without having carbon formation in the reaction zone. Conditions (area in Figure 7) to the left of the curve for various pressures show the zone at which carbon formation is favored. Temperature ranges to the right of each graph for a constant steam-to-methane ratio means operating on an area where carbon formation is not favored. Figure 7 shows that with increasing pressure, the carbon formation zone decreases. That is why for the large-scale hydrogen production by the steam methane reforming process, the pressure is above 10 bar.¹

5 | HEAT BALANCE

To determine the required heat for the SMR process, it is necessary to write the heat balance equation for the chemical system presented in Figure 1. The heat balance equation makes it possible to calculate the heat amount supplied per 1 mol of methane for the SMR process to achieve a targeted degree of the methane conversion. In general, the energy balance equation for a chemical-technological system in which work is not produced (not consumed) takes the form

$$\dot{Q} = - \sum_{i=1}^l \dot{n}_{i,j} \cdot H_i^{T_j}, \quad (28)$$

where $\dot{Q}$ is the rate of energy input to the reformer (J/s), $H_i^{T_j}$ is the molar enthalpy of the i th element at a temperature of T_j K (J/mol), l is the total number of inlets and outlets, the flows to the reformer is considered as positive, and the flows from the reformer are negative. For further analysis at the reformer inlet $j = 1$, at the reformer outlet $j = 2$.

Therefore, the heat balance equation for the reformer in Figure 1 can be written as

$$\begin{aligned} \dot{Q} = & \dot{n}_{H_2,2} \cdot H_{H_2}^{T_2} + \dot{n}_{CO,2} \cdot H_{CO}^{T_2} + \dot{n}_{CO_2,2} \cdot H_{CO_2}^{T_2} \\ & + \dot{n}_{CH_4,2} \cdot H_{CH_4}^{T_2} + \dot{n}_{H_2O,2} \cdot H_{H_2O}^{T_2} - \dot{n}_{H_2O,1} \cdot H_{H_2O}^{T_1} \\ & - \dot{n}_{CH_4,1} \cdot H_{CH_4}^{T_1}. \end{aligned} \quad (29)$$

FIGURE 6 Histogram of error distribution of the methane conversion: A, $T = 900$ K, $\beta=2$ (dashed line—Gaussian curve; solid line—histogram); B, fixed $T = 900$ K for $\beta = 1, \dots, 3$

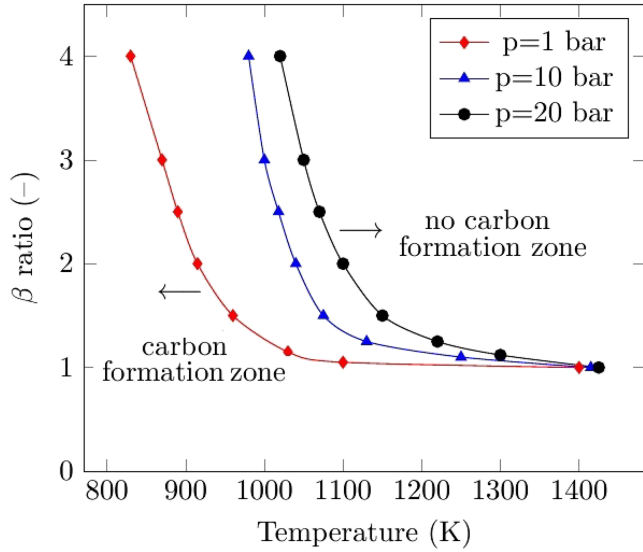
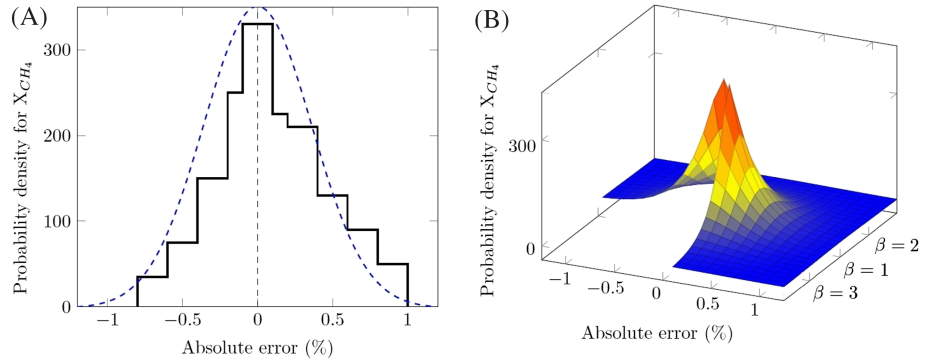


FIGURE 7 The operational conditions for the probable carbon formation in the steam methane reforming process

Using Equation (6) to (11), we can obtain the energy balance equation:

$$\begin{aligned} \dot{Q} = & \dot{n}_{CH_4,1}(3X_{CH_4} + X_{CO_2}) \cdot H_{H_2}^{T_2} \\ & + \dot{n}_{CH_4,1}(1 - X_{CH_4}) \cdot H_{CH_4}^{T_2} + \\ & + \dot{n}_{CH_4,1}(X_{CH_4} - X_{CO_2}) \cdot H_{CO}^{T_2} \\ & + \dot{n}_{CH_4,1}(\beta - X_{CH_4} - X_{CO_2}) \cdot H_{H_2O}^{T_2} + \\ & + \dot{n}_{CH_4,1}(X_{CO_2}) \cdot H_{CO_2}^{T_2} - \dot{n}_{CH_4,1} \cdot H_{CH_4}^{T_1} \\ & - \dot{n}_{CH_4,1} \cdot \beta \cdot H_{H_2O}^{T_1}. \end{aligned} \quad (30)$$

Equation (30) can be grouped as

$$\begin{aligned} \dot{Q} = & \dot{n}_{CH_4,1}[X_{CH_4}(3H_{H_2}^{T_2} + H_{CO}^{T_2} - H_{CH_4}^{T_2} - H_{H_2O}^{T_2}) \\ & + X_{CO_2}(H_{H_2}^{T_2} + H_{CO_2}^{T_2} - H_{CO}^{T_2} - H_{H_2O}^{T_2}) + \\ & + (H_{CH_4}^{T_2} - H_{CH_4}^{T_1}) + \beta(H_{H_2O}^{T_2} - H_{H_2O}^{T_1})]. \end{aligned} \quad (31)$$

And Equation (31) simplified as

$$\begin{aligned} \dot{Q} = & \dot{n}_{CH_4,1}[(H_{CH_4}^{T_2} - H_{CH_4}^{T_1}) + \beta(H_{H_2O}^{T_2} - H_{H_2O}^{T_1}) \\ & + X_{CH_4}(H_{smr}) + X_{CO_2}(H_{wgs})], \end{aligned} \quad (32)$$

where H_{smr} and H_{wgs} are the enthalpy of SMR reaction and WGS reaction, respectively (1 and 2 according to Table 1).

Equation (32) has a clear physical meaning where it is showing that the resulting transformation of substances in the steam methane reformer can be presented as a sequential heating of the feed streams of methane ($\dot{n}_{CH_4,1}$) and steam ($\beta \cdot \dot{n}_{CH_4,1}$) from the inlet temperature T_1 to the outlet temperature T_2 , the taking place of the SMR reaction at the temperature T_2 , the transformation of the part of the produced carbon monoxide (CO) via the WGS reaction at the temperature T_2 .

Equation (32) can be presented as

$$\begin{aligned} \psi = \frac{\dot{Q}}{\dot{n}_{CH_4,1}} = & (H_{CH_4}^{T_2} - H_{CH_4}^{T_1}) + \beta(H_{H_2O}^{T_2} - H_{H_2O}^{T_1}) \\ & + X_{CH_4}(H_{smr}) + X_{CO_2}(H_{wgs}), \end{aligned} \quad (33)$$

where ψ has the meaning of the heat amount supplied per 1 mol of methane entering the reformer.

Equation (33), taking into account the equations of the material balance, allows calculating the required heat flux to reach a targeted degree of methane conversion, which

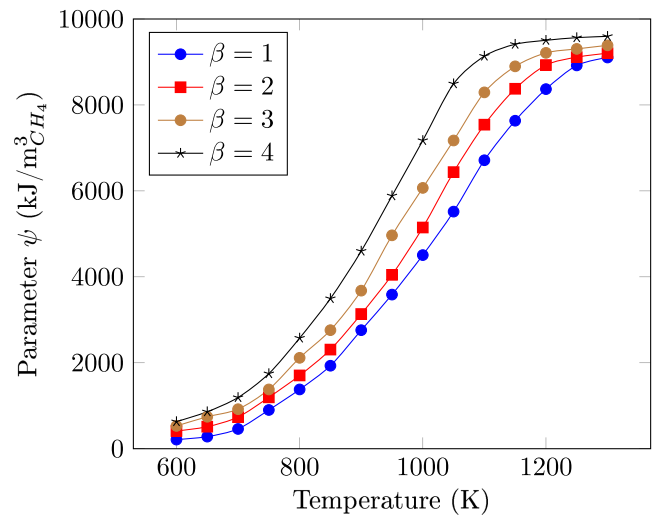


FIGURE 8 The heat amount supplied per 1 m³ (under n.c.) of methane entering the reformer for the equilibrium CH₄ conversion for the various β ratio at $p = 10$ bar

is less than or equal to the equilibrium value for any combination of parameters β , P , T_1 , and T_2 .

Figure 8 shows the dependence of ψ parameter as a function on the temperature for various steam-to-methane ratio at $p = 10$ bar. The inlet temperature is 600 K. As it can be seen from Figure 8, if the amount of steam increases in the inlet reaction mixture, then the amount of heat supplied for 1 m³ of methane increases. This is due to the fact that some of the heat is expended on steam heating from T_1 to T_2 , according to Equation (32).

6 | CONCLUSION

In this article, a thermodynamic analysis of the SMR is conducted in a wide range of operational conditions such as temperature, pressure, and the inlet steam-to-methane ratio. The results of thermodynamic analysis were compared with the results obtained via the different commercial and open-source programs such as Aspen-HYSYS, IVTANTHERMO, and Chemkin. All of these results are in excellent agreement.

The algorithm allows the technologist to establish an explicit dependence between the methane conversion in the reformer and operational parameters, ie, parameters that the technologist can intentionally change within certain limits to ensure the optimal process mode. Such parameters are the steam-to-methane ratio in the feed stream, the temperature of the mixture at the inlet to the reaction tubes, and the heat amount supplied to the reacting mixture through the walls of the reformer, and pressure. Similar to this algorithm can be used for the thermodynamic analysis of the other methane reforming processes, for example, the dry methane reforming process.

With help of developed algorithms, the operational conditions of the carbon formation were determined. It was established that for the temperature range above 1100 K, the probability of carbon formation is absent for steam-to-methane ratios above unit. Moreover, it was determined that with increasing pressure at the temperature range below 1100 K, the carbon formation zone decreases. That is why for the large-scale hydrogen production by the SMR process, the pressure is above 10 bar.

The heat balance equation for the steam methane reformer was written to determine the amount of heat supplied per 1 mol (m³ under n.c.) to the reformer in order to achieve a targeted degree of methane conversion. For this purpose, a parameter ψ was used. It was found that with an increase in the temperature, the amount of heat necessary to obtain an equilibrium degree of methane conversion increases. The obtained heat balance equation shows that the resulting transformation of

substances in the steam methane reformer can be presented as a sequential heating of the feed streams of methane ($\dot{n}_{CH_4,1}$) and steam ($\beta \cdot \dot{n}_{CH_4,1}$) from the inlet temperature T_1 to the outlet temperature T_2 , the taking place of SMR reaction at the temperature T_2 , the transformation of the part of the produced carbon monoxide (CO) via the WGS reaction at the temperature T_2 .

NOMENCLATURE

GFEM	Gibbs free energy minimization
G_{fi}^0	standard Gibbs free energy of i -element formation
$H_i^{T_j}$	molar enthalpy of the i th element at a temperature of T_j K
$H_{smr}; H_{wgs}$	the enthalpy of steam methane reforming reaction and water-gas shift reaction
$K_{eq,i}$	equilibrium constant of i -reaction i
$\dot{n}_{i,j}$	molar flow rate of i -element through j inlet (or outlet)
P	total pressure
P_i	partial pressure of i -element
$\dot{Q}$	rate of energy input to the reformer
R	molar gas constant
SMR	steam methane reforming
T	temperature
WGS	water-gas shift
x_i	molar fraction of i -element
X_{CH_4}	methane conversion
X_{CO_2}	methane conversion to CO ₂
β	steam-to-methane ratio
ΔG°	change of total Gibbs free energy
ψ	heat amount supplied per 1 mol of methane entering the reformer

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ORCID

Dmitry Pashchenko  <https://orcid.org/0000-0003-0249-6137>

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